

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA14

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO4: *Examine intermediate code generation techniques to enhance code optimization (BL4)*

Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:50

Annexure A

CLASS TEST

Code of Conduct:

I D K SUMALATHA(192424023) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Deepthi

Q. No	Understanding of Concepts (25)	Explanation (25)	Application (20)	Organization (15)	Accuracy(10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

Annexure A

CLASS TEST

Q1. Construct three address code for the following

if((a < b) and ((c < d) or (a > d))) then

z=x + y *z

else

z = z + 1

Q2.Translate (a < b) and (c < d) or (e < f) into three address statements using back patching.

Q3. Construct three address code using the following production for Boolean expression (a > b) or (c < d) or (e < f)

$B \rightarrow B1 \parallel B2$

$B \rightarrow B1 \ \&\& \ B2$

$B \rightarrow ! \ B1$

$B \rightarrow E1 \ \text{relop} \ E2$

$B \rightarrow \text{true}$

$B \rightarrow \text{false}$

Q4. Generate the three address code for the following program fragment-

while (A < C and B > D) do

if A = 1 then C = C + 1

else while A < = D do

A = A + B

18 Aug 26 CO4-AT class Test

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construct three address code for the following:

If $((a < b) \text{ and } (c < d) \text{ or } (a < d))$ then

$Z = X + Y * Z$

else

$Z = Z + 1$

Step-1: Understand the Boolean condition, $(a < b) \text{ AND } (c < d)$

the condition is

$a < b$

$c < d$

$a > b$

The structure is $(a < b) \text{ AND } [(c < d) \text{ OR } (a < d)]$

Step-2:- create labels L1, L2, L3, L4, L5, L6

Step-3:- Generate TAC for condition

L1:

If $a < b$ goto L2
goto L5

L2: If $c < d$ goto L4
goto L3

L3: If $a > d$ goto L4
goto L5

If $a < b$ is false, the complete AND expression is false, no control goes to the else part. If $a < b$ is true expression is evaluated. If either $c < d$ or $a > d$ is true, control goes to the then part.

Step-4:-

Generate TAC for the THEN part

The statement is $Z = X + Y * Z$. Multiplication has higher precedence.

30/40

L4:
 $t_1 = y * z$
 $t_2 = x + t_1$
 $z = t_2$
 goto L6
 Step-5 Generate TAC for the ELSE part
 L5:
 $t_3 = z + 1$
 $z = t_3$
 Final TAC for Q1

L1:
 if $a < b$ goto L2
 goto L5

L2:
 if $c < d$ goto L4
 goto L3

L3: if $a > d$ goto L4
 goto L5

L4:
 $t_1 = y * z$
 $t_2 = x + t_1$
 $z = t_2$
 goto L6

L5:
 L5
 $t_3 = z + 1$
 $z = t_3$

2) Translate $(a < b)$ and $(c < d)$ or $(e < f)$ into three address statements using backpatching.

given $(a < b)$ and $(c < d)$ or $(e < f)$

step-1: Apply operator precedence

AND has higher precedence than OR, so the expression is $[(a < b) \text{ AND } (c < d)] \text{ OR } (e < f)$

Step-2:-

Generate incomplete jumps

1. If $a < b$ goto -
2. goto -
3. If $c < d$ goto -
4. goto -
5. If $e < f$ goto -
6. goto -

The underscore represents a target address.

Step-3:-

Handle AND

For $(a < b) \text{ AND } (c < d)$, If $a < b$ is true, control must continue to the second condition.

Backpatch statement 1 to statement 3.

1. If $a < b$ goto 3

Step-4:-

Handle OR

If the first AND expression is false, the second OR operand $e < f$ must be evaluated.

Backpatch statement 2 to statement 5.

Backpatch statement 4 to statement 5.

Step-5:-

Final backpatched code.

1. If $a < b$ goto 3
2. goto 5

3. if $c < d$ goto 7
4. goto 5
5. if $e < f$ goto 7
6. goto 8
7. TRUE
8. FALSE

Backpatching means generating jump instructions first and filling target address.

2) construct three address code using the following production for Boolean expression $(a > b)$ or $(c < d)$ or $(e < f)$

$B \rightarrow B_1 \parallel B_2$

$B \rightarrow B_1 \text{ ~~and~~ } B_2$

$B \rightarrow !B_1$

$B \rightarrow E_1 \text{ relop } E_2$

$B \rightarrow \text{true}$

$B \rightarrow \text{false}$

Step-1:-

$[(a > b) \text{ OR } (c < d)] \text{ OR } (e < f)$

Step-2:

L₁: if $(a > b)$ goto L TRUE
goto L₂

Step-3:

check the 2nd condition.

L₂: if $c < d$ goto L TRUE
goto L₃

step-4: check the third condition.

L3: if $e < f$ goto LTRUE
if $e < f$ goto LTRUE
got Lfalse

final TAC for Q3

L1: if $a > b$ goto LTRUE
goto L2

→ Generate the three address code for the following program fragment while ($A < c$ and $B > D$) do if

$A = 1$ then $C = C + 1$
else while $A < D$ do
 $A = A + B$

complete TAC

L1: if $A < c$ goto L2
goto LEND

L2: if $B > D$ goto L3
goto LEND

L3: if $A = 1$ goto L4
goto L5

L4: $t_1 = C + 1$
 $C = t_1$
goto L1

L5: if $A \leq D$ goto L6

goto L1

L6:

$$t_2 = A + B$$

$$A = t_2$$

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	15%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps